School of Computing

Assessment Brief

Outline	Basic information
Module code and title:	COMP4104 Internet Technologies
Who to Contact Name & Email Address:	Yang He (y.he@kent.ac.uk)
Assessment name (if applicable):	Front-end web development mini-project
Assessment type and component:	Practical Competency
Assessment weighting:	60%
Module learning outcomes assessed in this assignment:	1,2,3,4
Submission deadline:	Tuesday, 17 Mar 2026 at 12:00 noon
Where to submit:	Moodle
How to request an extension:	Unexpected events such as illness, an accident, or bereavement sometimes happen. If so, you may need to request an extension, late submission or to be absent on the day of an assessed event. For instructions, see "in-course extenuating circumstance" in My Kent.
When marks and feedback on your submission will be provided on Kent Vision:	This should normally be three weeks after the deadline. For work submitted at the end of term 1 or 2, it will be the Friday of the first week of the following term.
How to find your feedback in Moodle	Your tutor's feedback on your assessment is one of the most valuable learning resources you have on your course. Please review it, compare it with feedback you've received on other assignments, and use it to identify skills where you can improve on future assignments. Your tutor, academic advisor, or a member of the Skills for Academic Success (SAS) team can help you apply the lessons in this feedback to future work.

1. The Task

1.1. What do I need to do?

In this assessment, you are expected to develop an interactive web application, "Matching Pairs Game", using HTML, CSS and JavaScript (JS). This game helps improve a player's memory by having them match pairs of cards based on their ranks.

Requirements and constraints

a) You should create a new folder named `a2-xyz`, and then create three text files in this folder: `xyz.html`, `xyz.css`, and `xyz.js` (*where `xyz` refers to your login)*
   - The CSS stylesheet for your webpage `xyz.html` should be defined in `xyz.css`
   - Your JS code for your webpage should be stored in `xyz.js`

b) You should download the file **`images.zip`** from Moodle and unzip the file. And then copy and paste the **`images`** folder into the folder you have created in (a)

c) Your web page should work as expected in Chrome and Edge web browsers.

General correctness

You are expected to follow the instructions as specified in the assessment. You should check that your HTML and CSS are valid, and your JS code contains no syntax errors.

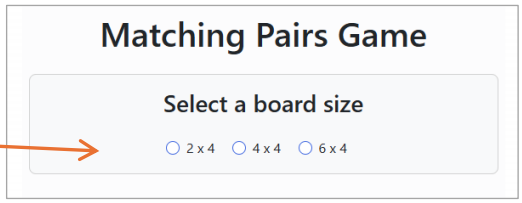
Design & Coding quality

- Your webpage should be user-friendly, with an appropriate layout, readable fonts, and a cohesive colour scheme.
- Your webpage should be responsive to various devices and accessible to a wide range of users.
- Your code should be well-structured, with appropriate comments and consistent indentation.
- Your JS code should be well-organised and free of unnecessary duplication. It should be easy for others to read, understand, and maintain.

Tasks

1. On loading the web document in a browser:
   - It should display a main heading and a brief instruction for players to start the game.
   - It should provide three radio buttons for players to select a board size: 2x4, 4x4 and 6x4.

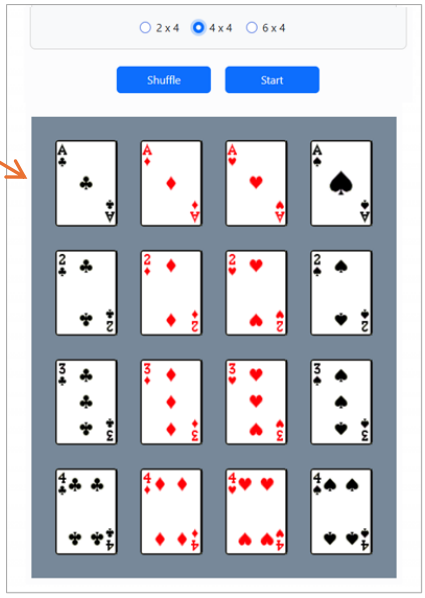
     Note: The game board should have 4 columns and a selected number of rows, 2, 4 or 6.



2. On selecting a board size, e.g. `4x4`:
   - It should have two buttons: "Shuffle" and "Start".

- It should display the game board consisting of 4 rows and 4 columns. Each column should contain cards from a single suit, arranged in ascending order of rank from 1 (i.e. Ace) to 4.

  The card image files are in the `images` *folder.*



3. Clicking on the "Shuffle" button should shuffle the cards randomly.

4. Clicking on the "Start" button:

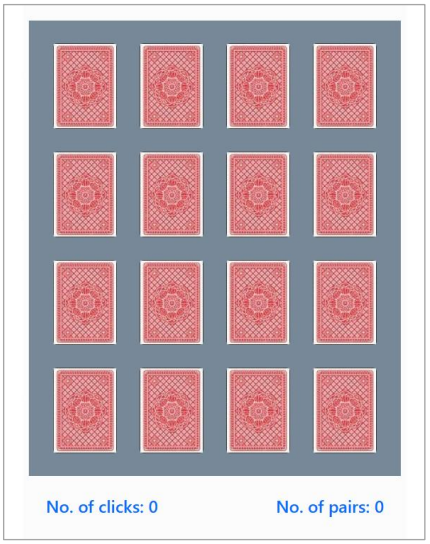
   If the cards have not been shuffled, it should pop up an alert box to instruct players to shuffle the cards before starting the game.

   Otherwise,

   - All cards on the game board should be flipped face down, i.e. in a hidden state.

     The back of the card image file `back.jpg` *is in the* `images` *folder.*

   - It should show the number of clicks and the number of pairs; both are 0 by default.
   - Both "Start" and "Shuffle" buttons, and all radio buttons, should no longer be accessible to players, e.g. disabled or hidden.

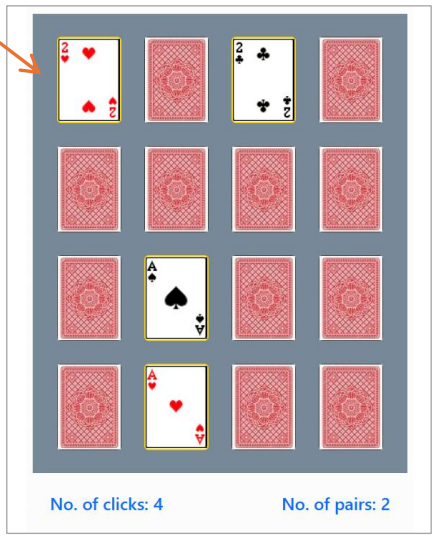


5. Clicking on a face-down card, it should flip the card face up, and the card becomes unclickable. The click counter should be updated accordingly.

   When clicking on another card, it should check whether the two selected cards form a pair, i.e. whether they have the same rank regardless of suit.

   If they are a pair, both cards should be highlighted in yellow. They should remain face-up and unclickable. The matching pair counter should be updated accordingly.

   Otherwise, both cards should be flipped face down again and remain available for selection.



6. The game is over when the player has matched all pairs of cards, or the number of clicks is more than twice the number of cards on the game board.

   When the game is over, it should pop up a dialog modal with a message and a "Play again" button.

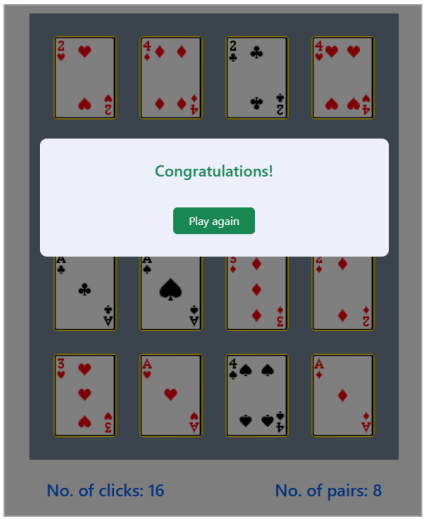
   - The message should be "Congratulations!" if the player has successfully matched all pairs of cards.

     Otherwise, it should be "Game over! Try again!"

     An example is shown on the next page.

- Clicking on the "Play again" button, it should restart the game as in Task 1.

  You may use `location.reload()` *to reload the web document.*



1.2. Where do I start? Tips for success.

Read the assessment brief – see **Section 1.1 What do I need to do?**
Start early: give yourself enough time to complete the assessment. Look at the lecture slides and the class materials. Read the relevant chapters of the textbook.

1.3. May I use Generative AI?

See the latest guidance at the My Kent page on Generative AI use in assessments.

1.4. What will I submit? Are there any special formatting or referencing styles expected?

You should compress the folder containing all your files for this assessment into a *single* ZIP file and then upload the .zip file using the link on Moodle *before* the deadline.

1.5. I have an Inclusive Learning Plan (ILP), how will this plan be accommodated?

In general, assessments have been designed to be inclusive. For timed assessments, extra time will automatically be given if your ILP recommends it. If you need more time for a coursework assessment due to the impact of your disability or mental health condition, please contact the student engagement team for your School to request an in-course extenuating circumstance. If you have an ILP that you think requires further accommodation, please contact the module convenor to discuss how it would work.

2. Support Available

2.1. Module activities or tasks that specifically help you prepare you for this assignment.

Lectures/classes in the first 4 weeks of term will provide the foundational tools and skills needed for this assignment. The class exercises / further study tasks set on Moodle during these weeks are also designed to help you prepare for this assignment.

2.2. Further assistance

For questions related to this assignment, you can contact Dr Yang He (y.he@kent.ac.uk).
See My Kent for links to support services and resources to help you with your studies. These include the Course Administration Team, Library Search and Subject Librarians, Skills for Academic Success, and Mental Health and Wellbeing Support.

3. Marking Criteria- How will my assignment be marked?

Criteria	Mark
General correctness	5
Design & Coding quality	5
Task 1	5
Task 2	6
Task 3	5
Task 4	6
Task 5	8
Task 6	10
Total	50 (100%)